04 / Implementation and QA

Architecture ownership, handoff sequence, source conflicts and evidence needed before any delivery is called implementation-ready.

STATUS IMPLEMENTATION PROTOCOL

Source: njr39/Cerebro-core at 1f78a1295d4dd36a11320995c1e14bf17788ac25. Earlier film-derived Lovart material is a separate reference. The PDFs are production instructions, not an asset approval or tested render.

Architecture observed in pinned code

FIELD	SOURCE-BOUND VALUE
Next / React / Three	16.4.2 / 19.3.0 / ^0.186.1
React Three Fiber	^9.8.2
Drei / Postprocessing	^10.7.9 / ^3.2.2
GSAP	3.13.0
Theatre.js	Not installed in pinned package; compatibility and migration proposed

scene.primary / src/cerebro/scene/CerebroScene.tsx

Own Canvas with CameraRig, SphereWorld and CinematicPostFX. active mount requires route trace

sphere.alternate / src/cerebro/sphere/CerebroSphere.tsx

Another Canvas with CameraRig; dynamically loaded from HqSphere. not proven simultaneously mounted; do not delete based on duplicate declaration

camera.rig / src/cerebro/camera/CameraRig.tsx

Camera control path; trace its actual mounts and inputs in Phase 01.

camera.motionOwner / src/cerebro/camera/motionOwner.ts

Owner arbitration for tour, voice, hands, touch and mouse.

sphere.world / src/cerebro/scene/SphereWorld.tsx

Renders scene including MediaCircleLayer.

media.circle / src/cerebro/scene/MediaCircleLayer.tsx

3D-linked HTML video and image preview candidates; inspect source and media streams before claims about fit.

material.shell / src/cerebro/appearance/surfaceMaterial.ts

Liquid glass preset: color #081018, transmission .98, roughness .03, metalness 0, IOR 1.52, thickness 2.6.

fx.shared / src/cerebro/scene/CinematicPostFX.tsx

Full-tier Bloom after threshold cap, SMAA, optional DOF/chroma/noise, Vignette.

composition.master / src/cerebro/sphere/masterComposition.ts

Projection fitter based on live stage and free area; FOV 52 for portrait aspect < .8, otherwise 46.

Ownership rule

One renderer lifecycle per intended viewport, one active camera controller, one writer per mutable property, and independently controlled components within that shared scene. The two Canvas declarations in source are not proof that both render simultaneously. Trace route mounts before deleting anything.

Explicit conflicts to resolve

ID	LEVEL	SOURCE RULE	OBSERVED / REQUEST
TYPE-01	HIGH	config.visualAuthority.lockedDecisions.fonts.titles = Sora and typography.families.display = Sora	src/app/globals.css --font-display starts with Jaapokki Subtract; CerebroSphere local UI specifies Plex.
COLOR-01	HIGH	config.visualAuthority.lockedDecisions.banned includes gold treatments	SOURCE_PALETTE.steel = #D4AF37 and cinematicOverride.fresnelRimHot references it. Prior film reference uses gold rails.
FX-01	HIGH	config.visualAuthority.lockedDecisions.banned includes glow and bloom	CinematicPostFX renders selective Bloom in full tier; cinematicFx defines bloom presets.
DURATION-01	MEDIUM	Latest repeated user request: three-minute film, 4K, continuous shot	Earlier cinematic JSON was 150 seconds; a prior output was 720p/180s; neither proves current source execution.
TIMING-01	MEDIUM	The approved motion tokens are source-bound and their statuses differ.	cinematicOverride proposes grainFps 24; a 180s 30fps social master requires 5400 output frames.
MEDIA-01	HIGH	User demands genuine Tarek/NOJARA media, fitted within designed placeholders, clear thumbnails.	Current procedural asset manifest does not verify portrait, showreel URLs, playable footage or publication rights.
MOTION-01	MEDIUM	V2 proposes Theatre.js authored keyframes.	The pinned package.json has R3F, Drei, GSAP, postprocessing, but no Theatre.js package.
PIXEL-01	HIGH	User asks pixel-indexed reusable assets and accurate placement.	Earlier 512×910 source-video maps are a different design reference. Runtime uses composeMasterFrame with container geometry; no current reference-to-render comparison is supplied.

Conflict handling

TYPE-01 / proposed resolution

Phase 01 identify exact routes and scopes; Phase 04 compare approved screenshots; retain source status until owner selects scope; no bulk font replacement.

COLOR-01 / proposed resolution

Keep reference-film gold separate from locked application identity; request explicit scoped approval before promoting any film-derived gold.

FX-01 / proposed resolution

Report render paths and screenshot delta; do not silently remove active FX or declare it approved.

DURATION-01 / proposed resolution

Use 180 seconds as the requested target for the proposed film build; preserve 150s legacy as historical, never merge timings.

TIMING-01 / proposed resolution

Treat grain cadence as an independently sampled texture, not the export frame rate.

MEDIA-01 / proposed resolution

Block media insertion until asset IDs, files, rights, duration, and poster frames are verified; no synthetic portfolio imagery.

MOTION-01 / proposed resolution

Propose integration only after version and ownership proof; existing camera owner retains authority until migration is tested.

PIXEL-01 / proposed resolution

Represent pixel bounds as null until approved reference dimensions, measured landmarks, and output captures exist; index computed responsive bounds separately.

Four sequential handoff gates

PHASE 01 / Engine and ownership

Scope: trace active mounts/routes, renderer/camera lifecycle, asset loaders, timing ownership, responsive sizing.

Exit evidence: build, typecheck, route startup, no verified duplicate writer. Stop before the next phase.

PHASE 02 / 3D scene, assets and film motion

Scope: approved GLB/texture/media audit, PBR and shader channels, lighting/postprocessing conflicts, 180-second authored path and frame graph, asset registry.

Exit evidence: visual comparison, camera continuity, media fit, provenance. Stop before the next phase.

PHASE 03 / Data, interaction and voice

Scope: real data contracts, input states, permission, single voice session, interaction handoffs.

Exit evidence: interaction tests, mute/interrupt, no synthetic voice state. Stop before the next phase.

PHASE 04 / Editorial overlays, capture and final QA

Scope: typography and RTL, social portrait/desktop layouts, visible logo, sound mix and master, 4K offline export where possible, full QC.

Exit evidence: frame scan 5400 frames, audio QC, responsive browser matrix, regression/performance evidence. Stop before the next phase.

Lovart versus coding agent

FIELD	SOURCE-BOUND VALUE
Lovart	Produce approved-source visual assets, editable layered scene files, fitted media surfaces, background passes, motion references and metadata.
Coding agent	Inspect active code, integrate exact approved assets, enforce component/camera ownership, wire runtime media and interaction, run tests.
Sound team	Record/clear VO and music, produce stems, mix/master measured output if Lovart has not been commissioned for audio.
Owner review	Decide font/gold/bloom scope, copy, logo/portrait source and final media rights before a visual variant is promoted to approved.

QA register and evidence

ID	METHOD	PASS CONDITION	NOW
DS-001	compare approved screenshots at identical viewport	measured design match	NOT STARTED
CAM-001	sample all 5400 camera frames, compare pose/velocity across ten joins	no discontinuity	NOT STARTED
MED-001	verify SHA-256, rights, posters and frame coverage for each media slot	all real source assets verified	BLOCKED
THUMB-001	inspect first-frame, middle-frame and cover at phone size	clear subject, undistorted frame, visible mark	NOT STARTED
AUD-001	measure full-duration integrated loudness and true peak of final mix	platform-approved target; no clipped peak	BLOCKED
RTL-001	capture real Arabic and mixed-language content	correct shaping, ordering, wrap and safe area	NOT STARTED
RESP-001	resize actual container, iPad split view, portrait/landscape and Safari/Chrome	no crop, overlap or camera reset	NOT STARTED
WEBGL-001	check texture color space, context loss, cleanup and frame timings	source-aligned materials and no persistent leak	NOT STARTED
EXPORT-001	inspect decoded 2160×3840 video frame count, sync, encoding and logo	5400 frames at 30fps, 180s audio, no black frames	BLOCKED

Per-output acceptance

- Output ID and filename match; source/derivative SHA-256 values recorded; alpha preview composites on navy, white, grey and checker.
- GLB imports with correct named hierarchy, pivot, bounding dimensions, normals, UV and PBR map channels; validate texture color spaces.
- Video fills only its declared surface with recorded fit/crop; poster is visually clear at 390px phone layout; source file and rights are verified.
- Camera and target paths sampled for all 5,400 frames. No unmotivated discontinuity at ten chapter joins.
- Editable text, official NOJARA logo and Arabic shaping are checked at portrait and landscape sizes; current locked app typography conflicts remain surfaced.
- A production claim requires decoded export, measured audio, real devices/browser captures, performance timings and owner sign-off.

Delivery status on 01 November 2026

These PDFs and the prior JSON package are prepared handoffs. They do not contain final Lovart renders, licensed portfolio media, completed 4K film, mastered audio, application implementation or production QA.